

A Polarization-Insensitive 3 dB Power Splitter Based On Inverse-Designed Subwavelength Pixelated Structure

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Abstract—The polarization-independent power splitters (PIPSs) are essential for enabling efficient integration in photonic circuits. In this letter, we propose and experimentally demonstrate an ultra-compact, broadband 3 dB PIPS, whose design is based on a sub-wavelength pixelated structure with a footprint of $2.5 \times 1.5 \mu\text{m}^2$. The initial pixelated structure is carefully designed to enhance algorithm robustness of the optimization process, thereby improving efficiency and avoiding local optima. The experimental results show that the insertion loss of both TE_0 and TM_0 is lower than 1.5 dB within the wavelength range of 1480–1580 nm, while the beam splitting ratio between the two output ports remains above 75%. In addition, the fabrication tolerance of the device is examined to ensure low insertion loss and reflection loss. This ultra-compact on-chip device demonstrates significant potential for advancing high-density integration in optical circuits.

Index Terms—Silicon photonics, integrated optical devices, power splitter.

I. INTRODUCTION

WITH the remarkable development of photonic integrated circuits (PICs) in recent years, the performance and functionality of systems-on-chip have been enhanced, and the innovation and evolution of these systems have been technically supported. Silicon-on-Insulator (SOI) has become one of the dominant platforms for building PICs due to its high compatibility with mature complementary metal oxide semiconductor (CMOS) processes [1], [2]. The high refractive index contrast of the platform offers several key advantages for PIC design, including compact size, low cost, and high integration density. As the optical field is highly confined

within the waveguide, it facilitates low-loss signal transmission and enable more precise optical manipulation. However, the high refractive index contrast increases the polarization dependence and affect device performance [3], especially in systems utilizing multiple polarization multiplexing.

Optical power splitters, as fundamental components in PICs, are used for distributing and routing optical signals, playing a key role in enhancing system performance and enabling high-density integration. Additionally, polarization-insensitive power splitters (PIPSs) on SOI platforms are in high demand. Over the years, several types of PIPSs have been reported on SOI platforms, including various schemes such as Y-branches [4], [5], multimode interference couplers (MMIs) [6], sub-wavelength gratings [7], [8], three-guide couplers [9], [10], [11] and directional couplers [12], [13]. In the compact PIPS fabricated on a 220 nm SOI platform, the Y-branch [4] achieves a loss of 0.25 dB for the TE mode and 0.23 dB for the TM mode. However, achieving polarization insensitivity requires a long device length ($14 \mu\text{m}$) due to the need for an adiabatic transition length. Subwavelength gratings [8] offer significant potential for compact power splitting ($\sim 4.3 \mu\text{m}$). However, practical fabrication may introduce substantial optical loss, limiting the bandwidth for 1×3 splitting to 54 nm for losses below 1.2 dB. The three-guide couplers [9] achieve TE and TM losses of 0.068 dB and 0.62 dB, respectively, with a coupling length of $7.3 \mu\text{m}$. However, it shows significant wavelength dependence during splitting, with the 1.5 dB bandwidth limited to 1528–1585 nm. The directional coupler [13] allows for achieving losses below 0.19 dB for both polarizations with a length of just $5 \mu\text{m}$, within a bandwidth of 1530 to 1600 nm. However, the device's minimum feature size is 30 nm, which presents a significant challenge for fabrication accuracy. Recently, inverse design has emerged as a highly attractive technique for developing devices with compact footprint and flexible functionalities [14], [15], [16], [17], [18]. This approach provides greater design freedom and offers a novel method for designing PIPSs.

In this letter, we propose and experimentally demonstrate an ultra-compact broadband PIPS optimized through inverse design. The device structure is created using algorithms optimized for a pixelated layout of only 25×15 pixels, with the pixel arrangement roughly determined prior to optimization to enhance algorithm robustness. The experimental results show that the fabricated device achieves an insertion loss

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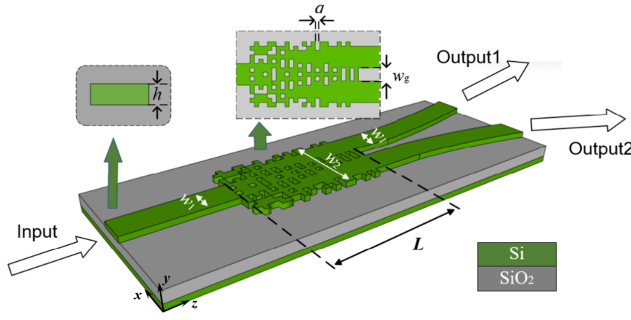


Fig. 1. 3D schematic diagram of the proposed 3 dB power splitter, together with top view of the pixelated area and cross-sectional view of the input strip waveguide.

(IL) of below 1.5 dB for both input polarized light within the wavelength range of 1480-1580 nm and maintains a splitting ratio (SR) exceeding 75% between the output ports. Compared to the aforementioned PIPS, our design focuses on high-density integration, achieving miniaturized footprint of just $2.5 \times 1.5 \mu\text{m}^2$ on the standard 220 nm SOI platform while ensuring broadband performance. The inverse-designed device exhibits significant potential for high-efficiency and high-stability designs in PICs.

II. DESIGN METHOD AND STRUCTURE

Fig. 1 illustrates the schematic structure of the proposed PIPS, including the 3D view of the device, the top view of the pixelated region, and the cross-sectional view of the input waveguide. The device consists of an input strip waveguide, two output bending waveguides, and an inverse-designed sub-wavelength pixelated structure. In the pixelated region, we engineered the spatial refractive index by etching nano-rectangular holes to achieve efficient beam splitting of the input polarized light. The region is composed of 25×15 pixels. The pixels are square with a side length of $a = 100$ nm, a setting chosen to accommodate the limitations of actual fabrication processes on feature size. The pixelated region has dimensions of $L = 2.5 \mu\text{m}$ and $w_2 = 1.5 \mu\text{m}$. The widths of the input and output ports w_1 , are set to 500 nm for C-band operation. The spacing w_g is chosen to be 300 nm to minimize interference from adjacent output ports. The device is designed on a standard SOI platform, featuring a 220 nm thick top silicon layer, a $2 \mu\text{m}$ buried oxide layer and a $2.2 \mu\text{m}$ SiO₂ upper cladding layer. The pixel structure is composed of alternating Si and SiO₂, allowing the device to be fabricated using a one-step etching process. In the simulation, all perfectly matched layer (PML) boundary conditions were selected to achieve balanced optimization for the performance of different input polarizations.

Considering that genetic algorithms require a long time for global search optimization and gradient-based algorithms have not yet been applied to polarization-insensitive optimization design, we prefer using the Direct Binary Search (DBS) algorithm to optimize the material distribution of pixels. To reduce iteration time, we prioritize a guided design of the initial pixel state. The specific steps are as follows:

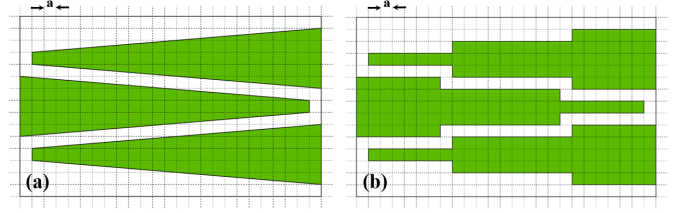


Fig. 2. (a) Schematic of the initial model consisting of a tapered waveguide; (b) Schematic of the initial model consisting of binarized pixels.

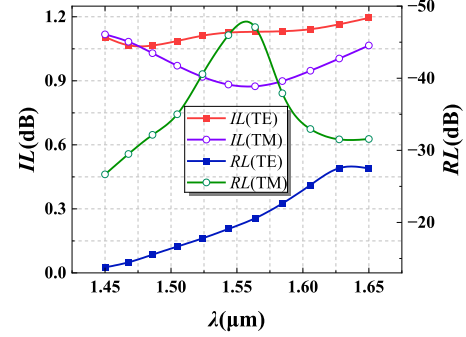


Fig. 3. Wavelength dependence of IL and RL for both input polarized modes in the binarized initial model.

(1) Constructing an initial structure using tapered waveguides to equally split the input polarized light, as shown in Fig. 2(a); (2) Approximate pixelization of each region based on its material occupancy, as shown in Fig. 2(b); (3) Iterative optimization using the DBS algorithm. The initial structure consists of one input waveguide and two output waveguides, with coupling length of $2.3 \mu\text{m}$ between them. The narrow end width of all tapered waveguides is selected as $0.1 \mu\text{m}$ through a simple parameter scan, resulting in a waveguide spacing of $0.1 \mu\text{m}$. During the optimization process, the region was kept axisymmetric to ensure an accurate 50:50 beam splitting ratio between the two output ports. The effective indices of Si and SiO₂ are 3.476 and 1.444, respectively. The device performance was determined and optimized using three-dimensional finite-difference time-domain (3D-FDTD) method calculations.

Fig. 3 shows the wavelength dependence of IL and reflection loss (RL) for both polarized lights input the binarized initial structure. Within the wavelength range of 1450-1650 nm, the initial binarized model exhibits ILs below 1.2 dB for both input polarizations, with RLs remaining less than -13 dB for TE and -26 dB for TM polarization, respectively. This structure ensures that the optimization algorithm starts from a promising initial point (shown in Fig. 3), thereby increasing the likelihood of finding the global optimum and enhancing the robustness of the algorithm. Next, the figure of merit (FOM) is calculated, defined as follows:

$$FOM = \gamma [\alpha (P_{\text{output1(TE)}} + P_{\text{output2(TE)}}) + \beta P_{\text{ref(TE)}}] + \delta [\alpha (P_{\text{output1(TM)}} + P_{\text{output2(TM)}}) + \beta P_{\text{ref(TM)}}] \quad (1)$$

Here, TE and TM represent the polarization states of the input modes. The coefficients α , β , γ , δ represent the weights

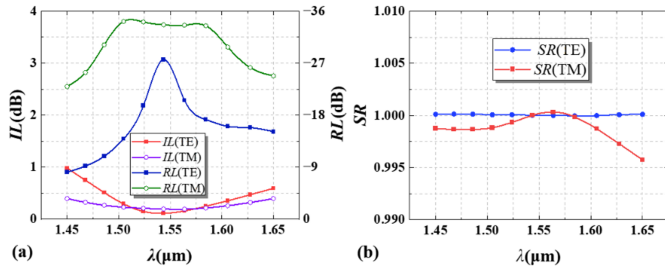


Fig. 4. (a) Wavelength dependence of proposed polarization-independent power splitter; (b) SR as a function of wavelength for both polarized modes.

assigned to the transmittance and reflectance for different polarization states, which can be dynamically adjusted during the optimization process. The material of the pixels is modified sequentially; if the FOM improves, the change is retained; otherwise, the modification is discarded, and the process continues to the next pixel.

III. SIMULATION RESULTS

Fig. 4 illustrates the performance of the algorithm-optimized PIPS, including the wavelength dependence of IL, RL and SR. The simulation results show that the ILs for both TE and TM polarizations are below 0.5 dB within the wavelength range of 1480–1625 nm, with the IL for TM polarization specifically below 0.5 dB across the operating bandwidth of 1450–1650 nm. At 1550 nm, the ILs are 0.12 dB for TE and 0.18 dB for TM, demonstrating the low loss and broad bandwidth of the devices. The RLs of TE and TM modes at 1550 nm are -26.5 dB and -33.2 dB, respectively. From 1450 nm to 1650 nm, the RL of TE mode exhibits significant wavelength dependence due to the electric field being perpendicular to the propagation direction, making it sensitive to pixel discontinuities. In contrast, the electric field of the TM mode is primarily in the propagation direction, making it less sensitive to lateral structural changes. The RL for TM mode remains consistently below -20 dB across the same range. Furthermore, the symmetric structure of the device ensures a 50:50 beam splitting ratio for both polarized lights, as shown in Fig. 4(b). Although the SR of TM shows relatively large deviations, the SR value with the largest deviation is 0.9957, which is still within the acceptable range for power splitting. When compared to the performance of the binarized initial structure shown in Fig. 3, the ILs of the algorithmically optimized pixelated structure have been significantly improved, while the SR is well-maintained, indicating the effectiveness of the optimization scheme.

Fig. 5 shows the simulated main field evolution for different polarized light inputs at 1550 nm, where the dotted box indicates the pixelated optimized region of the device. It can be observed that for both fundamental polarized lights launched from the input port, mode transition occurs within the pixelated region, causing the main power to be split into two beams, which are then routed through the bending waveguide.

To evaluate the robustness of the device against pixel size variations due to fabrication uncertainties, we investigated the impact of pixel side length offset Δa on the device performance, as shown in Fig. 6. The results indicate that the

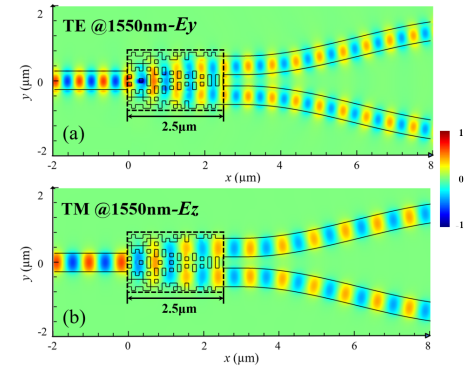


Fig. 5. Simulated main field evolution when (a) TE and (b) TM are input respectively.

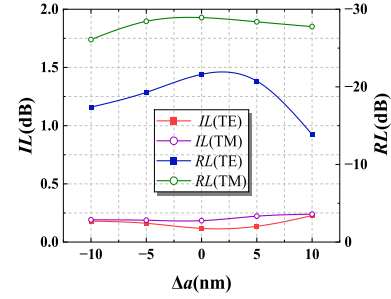


Fig. 6. Variation of device performance with pixel edge length offset Δa .

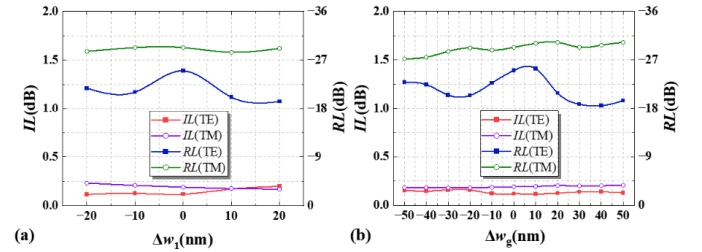


Fig. 7. Variation of device performance with: (a) input and output waveguide width offsets Δw_1 ; (b) output waveguide gap width offsets Δw_g .

ILs of the device all decrease slightly as the side lengths a of the rectangular pixels changes. At 1550 nm, the ILs of the two polarized light remains below 0.25 dB when the pixel side length varies by ± 10 nm. Additionally, the pixel size variation Δa has a relatively large effect on the RL, particularly for TE mode. When Δa ranges from $[-10, 8]$ nm, the RL for TE remains below -15 dB. This is due to the direct impact on the lateral electric field distribution or possible mode coupling in localized areas as the pixel size varies. For TM mode, The RL remains below -25 dB when Δa is within ± 10 nm.

The fabrication tolerances are also examined by varying the waveguide widths w_1 and the output waveguide spacing w_g , as shown in Fig. 7. Both offsets Δw_g and Δw_1 have minimal effect on the IL of the device. However, the RL for the TE mode shows more fluctuations. To ensure the RL remains below -20 dB for both polarization modes, Δw_1 and Δw_g should be controlled within ± 10 nm. Overall, the device performance at 1550 nm remains stable within ± 10 nm of fabrication error, indicating robust tolerance to fabrication variations.

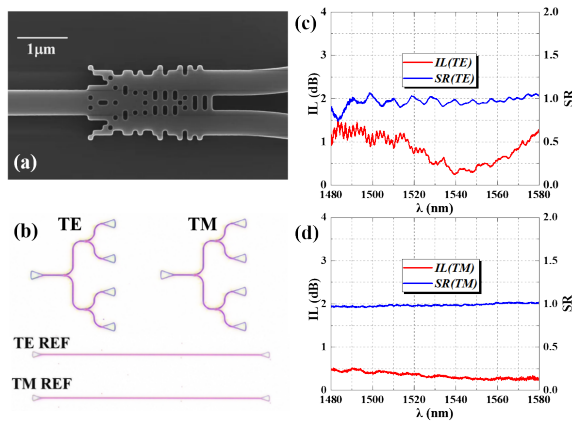


Fig. 8. (a) SEM image of the fabricated power splitter; (b) microscope image of the test configuration cascaded by the power splitter. The measured performance of the fabricated device when (c) TE-polarization and (d) TM-polarization are input respectively.

IV. FABRICATION AND CHARACTERIZATION

The proposed PIPS is fabricated on standard SOI wafers with a 220 nm thick top layer silicon and a 2 μm buried oxide layer. The fabrication process involves photoresist patterning using a 100 keV electron beam lithography, followed by etching the silicon waveguide structure with an inductively coupled plasma etching process. Finally, a 2.2 μm SiO_2 upper-cladding layer is deposited using plasma-enhanced chemical vapor deposition. The SEM image of the fabricated device is shown in Fig. 8 (a), revealing curved edges on the rectangular hole pixels due to the resolution limitations of the lithography process and diffraction effects. Fig. 8 (b) displays the waveguide configuration used for measuring the fabricated device. Fully etched grating couplers (GCs) with various structural parameters are employed for fiber-chip coupling in different polarization states, while reference waveguides for TE and TM modes are used to normalize the measured results. For the device under test, the output channels of the first-stage PIPS are cascaded with the input ports of the second-stage PIPS, and the output channels of the second-stage PIPS are used for characterization. The performance of the device is calculated using $\text{IL (dB)} = -10\log(P_{\text{ave-out}}/P_{\text{ref}})/2 + 3$ and $\text{SR} = P_{\text{min}}/P_{\text{max}}$. Here, $P_{\text{ave-out}}$ refers to the average optical power measured at the output ports, P_{ref} is the optical power measured at the reference waveguides, and P_{min} and P_{max} refer to the port powers corresponding to the minimum and maximum optical power detected at 1550 nm.

Fig. 8(c) and (d) shows the measured performance of the device. The experimental results reveal that for TE polarization, the IL remains below 1.5 dB within the wavelength range of 1480-1580 nm, though with considerable fluctuations, which are attributed to the curved edges of the rectangular pixels shown in Fig. 8(a). For TM polarization, the IL is less than 0.5 dB, with values of 0.44 dB for TE and 0.25 dB for TM at 1550 nm. Furthermore, the SR of the device is consistently maintained above 75%. It is important to note that the device's minimum feature size is 100 nm, which requires advanced etching techniques with high precision during fabrication. As manufacturing accuracy and photolithography resolution improve, the performance is expected to be further enhanced.

V. CONCLUSION

In summary, we have proposed and fabricated an ultra-compact and broadband 3dB PIPS. By modeling the initial pixelated structure before applying the DBS optimization algorithm, we enhanced the efficiency of the optimization process. The device was optimized within a 25×15 pixel area. Experimental results demonstrate that the device performs well with a stable beam splitting ratio. This work well demonstrates the advantage of using a guided initial model over a random model in the pixelated structural design step, thereby improving the robustness of the algorithm. Furthermore, with its ultra-compact size, the proposed PIPS has the potential to enhance the integration density of systems within PICs.

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